

DataSplits.jl: Data splitting strategies for machine learning in Julia

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Summary

DataSplits.jl is a Julia package (Bezanson et al., 2017) for constructing training, validation, test, and cross-validation partitions in scientific machine learning workflows. While random train/test splits and standard k-fold cross-validation are widely available in machine learning libraries, many scientific applications require more controlled splitting procedures. The package exposes a single entry point:

```
res = partition(data, strategy; kwargs...)
```

The returned split objects expose train, validation, and test indices, lazy observation views, materialised subsets, and row-pair representations suitable for external model-evaluation frameworks. Supported strategy families include representative sample selection for calibration modelling, distance-based selection in spectral or molecular descriptor spaces, group-aware validation to avoid leakage between related observations, and time-aware validation for temporally ordered data. The package is designed to be lightweight and interoperable with the Julia machine learning ecosystem, including MLUtils.jl, Flux.jl, and MLJ.jl. It can be installed from the Julia General Registry with `add DataSplits`. Source code is available at <https://github.com/davide-grheco/DataSplits.jl> and documentation at <https://davide-grheco.github.io/DataSplits.jl>.

Statement of need

The choice of data partitioning strategy can strongly affect the reliability of model evaluation, particularly in small-data scientific domains such as chemometrics, spectroscopy, metabolomics, materials informatics, and quantitative structure–property (QSPR) or structure–activity (QSAR) modelling. In these settings, independent and identically distributed (i.i.d.) random splits may be inappropriate: samples can be highly structured, class distributions may be imbalanced, repeated measurements may induce group leakage, and calibration sets may need to span the relevant feature and response spaces (Kohavi, 1995). Representative sample selection algorithms such as Kennard–Stone/CADEX (Kennard & Stone, 1969) and SPXY/MDKS (Galvão et al., 2005; Saptoro et al., 2012) are commonly used in chemometrics to construct calibration and validation sets with broad coverage of the experimental domain. Related approaches include the Duplex algorithm (Snee, 1977), which selects both training and test sets simultaneously using the maximin criterion; OptiSim and dissimilarity-based selection methods (Clark, 1997); the Onion method (Gallagher & O’Sullivan, 2020) and its joint X–y extension, XY-Onion (Ezenarro & others, 2025); electrostatic field-strength selection (He

et al., 2026); spectral clustering-based splits (Ng et al., 2001); and classification-oriented adaptations such as the Morais–Lima–Martin algorithm (Lelis Medeiros de Morais et al., 2019). Time-series and financial machine learning workflows call for chronological splitting strategies such as purged k-fold cross-validation and combinatorial purged cross-validation (López de Prado, 2018), and blocked cross-validation (Bergmeir & Benítez, 2012; Roberts et al., 2017). Target-aware cross-validation strategies such as Venetian blinds (Næs et al., 2002) assign samples sorted by response value across folds in round-robin order, guaranteeing uniform target coverage without binning. These methods are useful in scientific modelling but, to our knowledge, are not currently available together in any general-purpose Julia splitting library.

DataSplits.jl addresses this gap by collecting these splitting strategies in a single Julia package with a consistent interface. It is intended for researchers who need reproducible, inspectable, and domain-appropriate partitions while remaining compatible with existing Julia modelling tools.

State of the field

The closest Python equivalents are scikit-learn (Pedregosa et al., 2018) and astartes (Burns et al., 2023). Scikit-learn provides group-aware and time-series CV strategies (GroupKFold, TimeSeriesSplit, StratifiedGroupKFold) but does not implement distance-based selection methods such as Kennard–Stone, SPXY, or OptiSim. Astartes fills that gap for molecular and materials data, offering Kennard–Stone and several dissimilarity-based splitters, but does not cover group-aware CV, time-series CV, or nested cross-validation. DataSplits.jl covers both families: distance-based selection and a broad catalogue of standard CV strategies. On the samplers the two packages share, DataSplits.jl is 1.2 to 27 times faster than astartes and uses less peak memory (Table 1). Both packages precompute the full distance matrix for distance-based samplers, which limits the dataset size that fits in memory; DataSplits.jl also provides lazy variants that allow working with data of any size.

Table 1: Median runtime and peak memory of DataSplits.jl and astartes on identical synthetic inputs with 20 features. Scripts and results are in benchmark/experiment/.

Sampler	N	DataSplits.jl (ms)	astartes (ms)	Speed-up	DataSplits.jl (GiB)	astartes (GiB)
Kennard–Stone	10,000	389	1,126	2.9	0.75	1.58
SPXY	10,000	987	1,555	1.6	1.48	2.33
OptiSim	10,000	329	1,098	3.3	0.75	1.12
Sphere exclusion	10,000	545	676	1.2	0.75	1.49
Random	100,000	0.45	12.1	27	0.02	0.04
Target property (high)	100,000	2.9	56.2	19	0.02	0.05
Target property (low)	100,000	3.0	51.1	17	0.02	0.05
Time-based	100,000	6.7	21.1	3.2	0.02	0.04

Within Julia, MLUtils.jl provides an observation interface based on numobs, getobs, and lazy observation views, together with data loaders and basic splitting utilities (JuliaML, 2022). Flux.jl re-exports MLUtils data-loading functionality for neural network training (Innes, 2018). MLJ.jl provides a comprehensive model-evaluation framework with built-in holdout and cross-validation resampling strategies, and also accepts explicit train/test row-pair vectors for custom resampling workflows (Blaom et al., 2020). None of these three implement distance-based, group-stratified, or purged time-series splitting strategies. The closest Julia package in scope is ChemometricsTools.jl (Kneale, 2021), a broad chemometrics toolkit offering Kennard–Stone selection, two-way SplitByProportion and VenetianBlinds splits, and KFoldValidation/LeaveOneOut iterators, all returning materialised arrays. To our knowledge, no Julia package implements group-stratified or purged time-series splitting.

77 DataSplits.jl is not intended to replace these packages. Instead, it complements them
78 by focusing on splitting strategies that are especially useful in scientific datasets and
79 calibration/validation design. The output of these strategies can be consumed by other
80 packages: lazy views can be passed to Flux/MLUtils data loaders, while explicit row pairs can
81 be supplied to MLJ evaluation routines without requiring DataSplits.jl to depend on MLJ.jl.

82 DataSplits.jl was developed as a separate package rather than an extension to MLJ.jl or
83 MLUtils.jl because its core contribution is split construction rather than model evaluation or
84 data loading. Keeping the package lightweight allows the same split definitions to be reused
85 across MLJ, Flux, MLUtils, and custom scientific workflows without imposing a dependency
86 on any specific modelling framework.

87 Software design

88 The central abstraction in DataSplits.jl is a splitting strategy. Each strategy is represented
89 by a concrete subtype of `AbstractSplitStrategy`, while cross-validation strategies subtype
90 `AbstractCVStrategy`. Users call `partition(data, strategy; kwargs...)` to produce an
91 `AbstractSplitResult`. A single entry point is preferable to separate per-algorithm functions
92 because it lets user code switch strategies without changing call sites. Result types include
93 two-way train/test splits, three-way train/validation/test splits, and cross-validation splits
94 containing one split per fold.

95 The package exposes stable accessors such as `trainindices`, `testindices`, `valindices`, and
96 `folds`, so downstream code does not need to depend on internal result fields. Exposing indices
97 rather than only data subsets matters in practice: the same split can be applied to multiple
98 aligned arrays (features, labels, metadata, auxiliary covariates) without re-running the selection
99 algorithm. For data extraction, DataSplits.jl provides both lazy and materialised interfaces.
100 Functions such as `trainview`, `testview`, `valview`, and `splitview` use MLUtils-compatible
101 observation views, while `traindata`, `testdata`, `valdata`, and `splitdata` materialise subsets
102 using `getobs`.

103 All distance-based strategies have a lazy counterpart (e.g. `LazyKennardStoneSplit`,
104 `LazyOptiSimSplit`) that computes distances on-the-fly in $O(N)$ peak memory instead of
105 precomputing the full $O(N^2)$ distance matrix. This makes the strategies applicable to datasets
106 where the full matrix would not fit in RAM, at the cost of increased runtime — a trade-off
107 that is important for the high-dimensional descriptor matrices common in chemometrics and
108 molecular modelling. At $N = 10,000$ the lazy variants allocate 236 KiB where the eager ones
109 allocate 763 MiB, for a 1.4 to 2.4 times increase in runtime. Under a 4 GiB budget this raises
110 the largest feasible sample count from 10,000 to at least 100,000.

111 The package is designed to be extensible. Adding a custom strategy requires subtyping
112 `AbstractSplitStrategy`, declaring which auxiliary slots (`:data`, `:target`, `:time`, `:groups`)
113 the strategy reads via the `consumes` trait, and implementing a single `_partition` method.
114 The `partition` entry point handles cohort-size resolution, slot validation, and feature-matrix
115 conversion automatically.

116 The time-aware strategies include `BlockedCV`, which implements blocked cross-validation
117 (Bergmeir & Benítez, 2012) by partitioning the time axis into contiguous blocks to avoid
118 look-ahead bias, and `PurgedKFold`, which implements purged k-fold cross-validation (López de
119 Prado, 2018) by additionally removing observations from the training fold whose labels overlap
120 in time with the test fold.

121 A key design choice is to keep ecosystem integration lightweight. DataSplits.jl provides
122 `rowpairs`, which converts any split result into the vector of `(train_indices, test_indices)`
123 pairs accepted by MLJ's `evaluate!` `resampling=` keyword, without requiring a dependency
124 on MLJ.jl or MLJBase.jl. This avoids coupling split construction to any particular modelling
125 framework.

126 The test suite checks split-size resolution, non-overlap between cohorts, complete fold coverage
 127 where applicable, reproducibility under fixed random number generators, stratification and
 128 group constraints, and consistency of lazy and materialised data extraction.

129 Example usage

130 A representative train/test split can be constructed using the Kennard–Stone strategy:

```
using DataSplits, Random
```

```
X = randn(Xoshiro(42), 8, 200)
y = vec(sum(X; dims = 1)) .+ randn(Xoshiro(7), 200)

res = partition(X, KennardStoneSplit(); train = 0.8, test = 0.2)

X_train, X_test = splitview(res, X)
train_idx = trainindices(res)
test_idx = testindices(res)
```

131 For stratified cross-validation with MLJ, the result can be converted to row pairs directly. MLJ
 132 tables place observations in rows, so the feature matrix is transposed first:

```
using MLJ, DataFrames
```

```
Xt = DataFrame(permutedims(X), :auto) # 200 rows x 8 features
labels = coerce(ifelse.(y .> 0, "high", "low"), Multiclass)

cvs = partition(Xt, StratifiedKFold(5); target = labels, rng = Xoshiro(42))
model = ConstantClassifier() # any MLJ model
mach = machine(model, Xt, labels)

evaluate!(mach; resampling = rowpairs(cvs), measure = accuracy)
```

133 The same split indices can also be applied to multiple aligned data sources, enabling integration
 134 with Flux data loaders, which take observations along the last dimension as DataSplits does:

```
using Flux
```

```
model = Flux.Dense(8 => 1)
loader = Flux.DataLoader(trainview(res, X, y); batchsize = 64, shuffle = true)
```

135 In chemometric calibration, Kennard–Stone-style selection constructs a training set that spans
 136 the observed descriptor space, reducing the risk that validation samples fall outside the
 137 calibration domain — a problem that random splits cannot guard against by design. The
 138 practical difference is illustrated in [Figure 1](#).

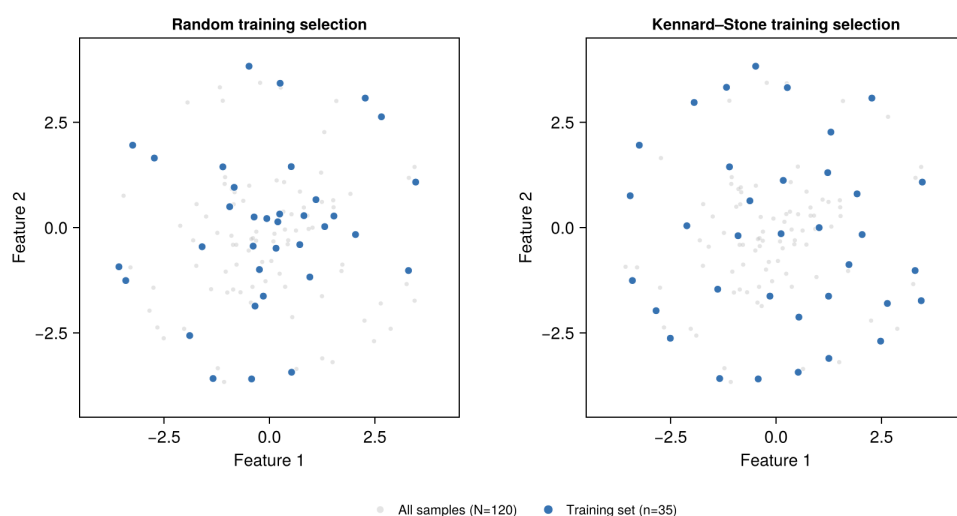


Figure 1: Comparison of random and Kennard–Stone training-set selection on a synthetic dataset with a dense central region and a sparse outer boundary. Gray points denote all samples ($N=120$) and blue points the selected training set ($n=35$). Random selection follows the empirical density and concentrates training samples in the center, whereas Kennard–Stone produces a more space-filling selection that better covers the boundary of the experimental domain.

Strategy family	Examples	Typical use
Distance-based	KennardStoneSplit, SPXYSplit, OptiSimSplit, DuplexSplit, OnionSplit, FieldStrengthSplit, SpectralSplit	Calibration/test design in chemometrics and descriptor spaces
Target-aware	TargetPropertySplit, StratifiedKFold, StratifiedShuffleSplit, XYOnionSplit, VenetianBlindsCV	Class-balanced or response-aware validation
Group-aware	GroupShuffleSplit, GroupKFold, StratifiedGroupKFold	Avoiding leakage across related observations
Time-aware	TimeSplit, BlockedCV, PurgedKFold, CombinatorialPurgedKFold	Temporally ordered validation
Resampling	BootstrapSplit, ShuffleSplit, RepeatedKFold, NestedCV	Repeated performance estimation and hyperparameter tuning

Research impact statement

DataSplits.jl has been developed to support benchmarking and validation workflows in scientific machine learning where split construction directly affects reported performance estimates. The package underpins a published companion study (Crucitti & Mosquera Orgueira, 2026), which demonstrates empirically that split strategy choice can dominate differences between modelling algorithms in structured chemical datasets, with random splits systematically producing more optimistic performance estimates than distance-based and group-aware alternatives. A reproducible tutorial (notebooks/tutorial.jl) included in the repository provides side-by-side comparisons of random, Kennard–Stone, SPXY, and group-aware splits on controlled datasets, illustrating how coverage of feature and response spaces differs across strategies and quantifying group leakage under random versus group-aware partitioning. The tutorial also

generates Figure 1. A benchmark suite (benchmark/benchmarks.jl) tracks runtime scaling of all strategies, including lazy variants, to support informed strategy selection on datasets of varying size. DataSplits.jl will be maintained by the GrHeCo-Xen group through the public GitHub repository, where users can report issues, request features, and contribute new splitting strategies.

AI usage disclosure

Generative AI tools, including Claude Code (Anthropic), were used to assist with software development — specifically in the integration of some algorithms — documentation drafting, and drafting and language editing of this paper. AI tools were not used to make autonomous design, implementation, or scientific decisions. All AI-assisted code was thoroughly verified against independent reference implementations and covered by the test suite. All AI-assisted text was reviewed and edited by the authors, and technical claims were checked against the implementation, tests, documentation, and cited literature. The authors are responsible for the final content of the manuscript, code, and documentation.

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